Engineering Reboot 2026

Day 24 Engineering Workbook

Version 1.0 – Implementation Edition (Locked)

Mission

Extract useful business information from collections by searching, filtering, sorting, and summarizing the Employee Directory dataset.

Learning Objectives

Search collections efficiently.

Filter data using business rules.

Sort collections with Python tools.

Compute summary statistics.

Write reusable collection-processing functions.

Engineering Theory

Searching locates records. Filtering creates subsets. Sorting changes presentation. Aggregation summarizes data.

Week 5 Engineering Scenario

Extend the Employee Directory application by adding reporting capabilities while continuing to use the canonical EMPLOYEES dataset.

Reference Dataset

Reuse employee_data.py from Day 23 without modification.

Exercise 1

search_demo.py: Search employees by ID and by name.

Exercise 2

filter_demo.py: Display Engineering employees, employees earning over $80,000, and employees older than 35.

Exercise 3

sort_demo.py: Sort employees by name, salary, and age.

Mini Project

Extend the Employee Directory into an Employee Reporting System with menu options for search, reporting, and summary statistics.

Stretch Goals

Total payroll; employee count per department; salary range search; case-insensitive search.

Engineering Checklist

Single-responsibility functions; meaningful names; reusable dataset; no exceptions.

Git Commit

Day 24 - Completed employee search, filtering and reporting system

Engineering Takeaways

Understand searching vs filtering, sorting, aggregation, and reusable processing functions.